Overcoming Data Limitations in Credit Risk Assessment with FinGPT-Generated Synthetic Data

Yuanxi Cai
Automation
Tongji University
Shanghai, China
22252253@tongji.edu.cn

***Abstract*—In credit risk management, accurately assessing credit risk is critical for the stability and risk mitigation of financial institutions. However, research in this area is often hindered by the limitations of small and imbalanced datasets, which impair the effectiveness of credit classification models. This study addresses the challenge of insufficient financial data by utilizing FinGPT, an open-source large language model, to generate synthetic datasets for credit risk assessment. The study compares the performance of three traditional machine learning algorithms, namely Random Forest, Decision Tree, and K-Nearest Neighbors, trained on both the original and FinGPT-generated datasets. The experimental design involves splitting the dataset into training and testing sets, with different portions used to simulate scenarios of limited data availability. FinGPT, which does not require additional fine-tuning, is evaluated against these traditional models using key metrics including Accuracy, Recall, Precision, and F1 Score. The findings demonstrate that FinGPT can enhance the predictive accuracy of these models, particularly in scenarios where data is scarce, highlighting its potential as a powerful tool for financial applications. This study effectively explored a new possibility in overcoming the limitations associated with small or imbalanced datasets based on the value of synthetic data.**

Keywords*—*FinGPT, credit risk management, machine learning

I. Introduction

In the realm of credit risk management, the accuracy of credit risk assessment stands as a cornerstone, intimately tied to the risk mitigation capabilities and operational stability of financial institutions. However, research endeavors in this domain are frequently constrained by the idiosyncratic characteristics of datasets, particularly their small sample sizes and imbalanced nature, which notably impair the efficacy of credit classification models. The inherent confidentiality surrounding financial transactions further complicates data acquisition, resulting in a scarcity of publicly available, high-quality datasets. This predicament is particularly acute for nascent financial institutions or entities venturing into novel credit services, who grapple with high costs and complexities in data procurement, thereby exacerbating the small sample size issue. Small sample data, characterized by incompleteness and inadequacy, often lead to the development of inefficient or even ineffective models, underpinning flawed decision-making processes [1-7]. This can lead to serious consequences, such as credit fraud.

To address the issue of insufficient financial data in the early stages, both academia and industry have extensively explored machine learning techniques to mitigate challenges posed by small sample sizes. For example, Zhang [8] employed extreme learning machines to synthesize samples and implemented feature engineering to alleviate attribute scarcity. However, these approaches remain grounded in relatively basic supervised learning methods. Meanwhile, Lopez-Rojas et al. demonstrate the application of PaySim [3], a financial simulator that mimics mobile money transactions based on original datasets, for fraud detection data generation [9]. By employing agent-based simulation techniques and mathematical statistics, they affirm that simulated data can be as prudent as original datasets for research purposes [10]. However, these methods largely overlook the unique complexities and dynamics of the financial sector, failing to offer tailored solutions. Consequently, existing research still faces significant limitations in enhancing the generalization capabilities and prediction accuracy of credit risk assessment models.

To address the research gap, this study proposes an innovative solution centered on the use of the open-source large language model FinGPT [4] to generate synthetic datasets for supporting credit risk assessment research. By utilizing its powerful language understanding and generation capabilities, FinGPT extracts and integrates key information from proprietary datasets to create synthetic datasets that not only reflect real-world trading environments but also include a variety of credit scenarios [11]. This solution possesses inherent learning and predictive capabilities, and compared to traditional algorithms, its predictive accuracy is significantly improved. Additionally, this study not only presents the experimental results of three machine learning algorithms applied to credit assessment tasks but also provides an analysis of different performance metrics.

II. Method

This section outlines the methodology employed throughout the experimental procedure, detailing each phase from data importation to the final evaluation. The experimental design comprises two primary components: Large Language Models (LLMs) and traditional machine learning algorithms. To ensure a comprehensive assessment of the model's performance, four key metrics were selected for evaluation. Each step of the process is systematically described, providing clarity on the implementation and rationale behind the chosen methods. This structured approach ensures the robustness and reliability of the experimental outcomes.

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A. Overview of the Model Workflow

As shown in Fig. 1, the experimental framework in this study has three main blocks- and several key steps: data collection, data preprocessing, dataset construction, LLM experiment, ML experiment, and evaluation.

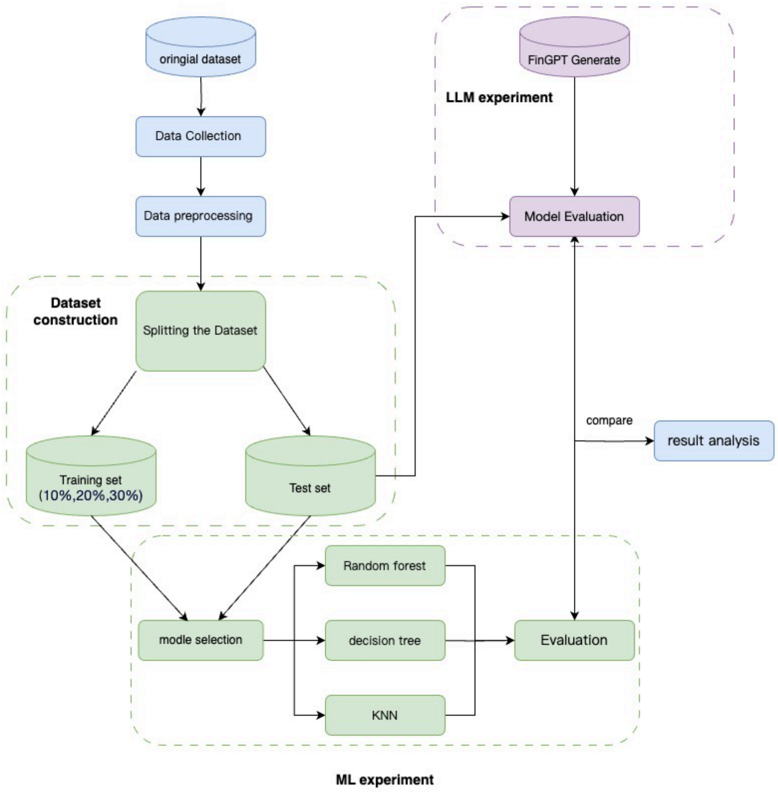

Fig. 1. General flowchart of credit risk assessment in this study.

First, the original dataset is collected and preprocessed for analysis. The dataset is then split into a training set and a test set, with the training set further divided into different sizes (i.e. 10%, 20%, 30%) to simulate the scenarios of limited samples and evaluate model performance. The framework includes two parallel processes. One involves training the FinGPT model, which is then evaluated. The other process focuses on traditional models: Random Forest [12], Decision Tree [13], and K-Nearest Neighbors (KNN) [14]. These models are trained with the prepared data and evaluated for accuracy and other metrics. Finally, the results from both FinGPT and traditional models are analyzed and compared in the result analysis stage. This thorough approach provides insights into the effectiveness of different models for credit risk classification.

B. Datasets

This project includes two notable datasets: the Australian Credit Approval dataset and the German Credit Data dataset [15]. Both datasets are derived from real-world financial contexts and are used extensively in academic research, particularly in the domain of credit risk analysis and the evaluation of machine learning algorithms. These datasets provide a diverse set of attributes and challenges, including missing values, categorical data, and cost-sensitive classification, making them valuable tools for developing and testing predictive models.

TABLE I. ATTRIBUTE LIST SUMMARY FOR TWO CREDIT DATASETS

Attribute No	Attribute	Value
1	Case No	Number (1-690 for Australian, 1-1000 for German)
2-6	Continuous Attributes	Number (Varies based on dataset)
7-12	Nominal Attributes	Binary (0, 1) or Categorical (Various strings)
13	Age	Number (Australian dataset only)
14	Credit Score	Number (German dataset only)
15	Ethnicity	String (German dataset only)
16	Jaundice	Boolean (True or False)
17	Family mem with ASD	Boolean (True or False)
18	Who completed the test	String
19	Class/Credit Risk	Binary (1: Good, 2: Bad for German, 0 or 1 for Australian)

Table I summarizes the detailed description of the dataset. The name of the dataset is “Credit Datasets”, which contains a total of 1690 records across two distinct datasets: the Australian Credit Approval dataset and the German Credit Data dataset. These datasets include 19 variables that represent various features associated with credit risk assessment. Among these variables, ten are used to determine specific attributes such as continuous or nominal values, as denoted by items 2 through 12 in the table above. These variables are essential in evaluating the creditworthiness of individuals.

In particular, the German Credit Data dataset includes a clinically significant variable known as the Credit Score, which is crucial for determining an individual's credit risk. The individuals in the German dataset are classified as "Good" or "Bad" based on their credit score, with higher values indicating a more favorable credit status.

Table II in the report shows a sample of 10 records in comma-separated values (CSV) format of the Credit Datasets, illustrating the structure and types of data included in the dataset.

TABLE II. A SAMPLE OF 10 RECORDS IN THE COLLECTED DATASET.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	22.08	11.46	2	4	4	1.585	0	0	0	1	2	100	22.08	11.46
2	22.67	7	2	8	4	0.165	0	0	0	0	2	160	22.67	7
3	29.58	1.75	1	4	4	1.25	0	0	0	1	2	280	29.58	1.75

C. FinGPT

FinGPT is a versatile, data-efficient and training-free synthesis data method built upon LLMs. The development of FinGPT stems from the growing interest in applying LLMs to finance, where proprietary models like BloombergGPT have dominated but remain inaccessible due to their high cost and exclusive data access.

FinGPT adopts a data-centric approach, emphasizing the importance of data acquisition, cleaning, and preprocessing. It is structured as an end-to-end framework with four core layers: Data Source, Data Engineering, LLMs, and Applications. This structure ensures comprehensive market coverage, real-time processing, and adaptability to the dynamic nature of financial data.

FinGPT is not just a technical tool but also a community-driven initiative under the AI4Finance Foundation, aiming to foster innovation and collaboration in open finance. By offering accessible resources, FinGPT enables the development of customized FinLLMs for various financial applications, such as robo-advising and algorithmic trading, promoting the broader adoption of AI in finance.

D. Machine Learning Models-based Prediction

To evaluate the effectiveness of different classifiers in predicting credit risk, this paper utilized a high-quality dataset generated by FinGPT to train the models, comparing their performance against those trained on the original dataset. This project also aims to compare the numerical outputs generated by the FinGPT model with those produced by machine learning models trained directly on the original dataset, as well as with established baseline machine learning models. The machine learning baselines are as follows:

- **Random Forest**, which constructs multiple decision trees and aggregates their outputs to enhance accuracy and reduce overfitting [16], using 100 trees for optimal balance.
- **Decision Tree** classifiers, which use a tree-like model based on the Gini impurity criterion [17], provide an easily interpretable structure by recursively splitting the data into homogenous subsets.
- **K-Nearest Neighbors**, a non-parametric algorithm that classifies based on the majority class of the nearest five neighbors [18], making it effective for complex feature relationships.

All models were applied to the standardized Australian credit dataset to ensure a fair and comprehensive performance evaluation.

E. Implementation Details

This study conducted a series of experiments using three classification algorithms: Random Forest, Decision Tree, and K-Nearest Neighbors. The experiments aimed to compare the performance of these models on the Australian credit dataset, with particular focus on key metrics such as Accuracy, Recall, F1 scores and Precision. The data was standardized before modeling to ensure consistency and comparability across models.

For the Random Forest model, the number of trees was set to 100, ensuring a balanced trade-off between computational efficiency and model performance. The Decision Tree classifier, using the Gini impurity criterion, served as a baseline for comparison, while the KNN classifier was configured with k=5 neighbors, reflecting a commonly used setting in classification tasks.

Later, FinGPT was able to generate high-quality financial data with zero samples, which was then provided to traditional models for training. This approach can enhance model performance, demonstrating the effectiveness of the FinGPT architecture in handling complex financial data generation tasks.

III. RESULTS AND DISCUSSION

Table III presents the results of training three fundamental machine learning algorithms—Random Forest, Decision Tree, and K-Nearest Neighbors (KNN)—on the original dataset (10%, 20%, 30%) to predict credit risk assessment. This table highlights how these models perform when applied to actual, unaltered data. On the other hand, Table IV shows the performance of the same three algorithms, but this time trained on a dataset generated using the FinGPT tool. By utilizing the generated data, the goal is to observe how these models adapt and perform under potentially more diverse or synthesized conditions, which might differ from the original dataset in terms of feature distribution and variety. The comparison between these two tables allows for an analysis of how well the algorithms generalize and perform when trained on authentic data versus data generated by FinGPT, providing insights into the robustness and flexibility of the models in different data environments.

TABLE III. PERFORMANCE COMPARISON OF DIFFERENT ML MODELS BASED ON ORIGINAL DATASET.

	Accuracy ↑	Recall ↑	Precision ↑	F1 Score ↑
Random Forest (10%)	0.8535	0.7625	0.8235	0.7918
Random Forest (20%)	0.8659	0.8265	0.8413	0.8339
Random Forest (30%)	0.8654	0.8662	0.8354	0.8504
Decision Tree (10%)	0.8100	0.7000	0.7350	0.7170
Decision Tree (20%)	0.7790	0.8030	0.7740	0.7882
Decision Tree (30%)	0.8137	0.7250	0.7644	0.7441
KNN (10%)	0.8148	0.7205	0.7789	0.7485
KNN (20%)	0.8351	0.8130	0.8014	0.8072
KNN (30%)	0.7992	0.7345	0.7668	0.7504

TABLE IV. PERFORMANCE COMPARISON OF DIFFERENT MODELS BASED ON GENERATE DATASET.

	Accuracy ↑	Recall ↑	Precision ↑	F1 Score ↑
Random Forest (10%)	0.8689	0.7800	0.8455	0.8120
Random Forest (20%)	0.8899	0.8430	0.8733	0.8580
Random Forest (30%)	0.8903	0.8850	0.8584	0.8713
Decision Tree (10%)	0.8457	0.7680	0.7690	0.7685
Decision Tree (20%)	0.7988	0.8450	0.7990	0.8211
Decision Tree (30%)	0.8355	0.7780	0.7844	0.7812
KNN (10%)	0.8254	0.7695	0.7933	0.7812
KNN (20%)	0.8598	0.8470	0.8243	0.8355
KNN (30%)	0.8179	0.7875	0.7982	0.7928

A. The Performance of Models

The results presented in Table III showcase the Random Forest model demonstrates superior performance across all metrics, particularly with higher data percentages. For instance, with 30% of the data, Random Forest achieves an Accuracy of 0.8654, Recall of 0.8662, Precision of 0.8354, and an F1 Score of 0.8504, indicating a robust predictive capability.

In contrast, Table IV reflects the performance of the same algorithms trained on a dataset generated using the FinGPT tool. The generated data appears to enhance model performance, especially for Random Forest, which shows further improvements across all metrics. For example, with 30% of the generated data, the Random Forest model achieves an Accuracy of 0.8903, Recall of 0.8850, Precision of 0.8584, and an F1 Score of 0.8713. These results suggest that the generated data might provide better generalization, as indicated by the increased metrics across the board.

B. Discussion of the Model Performance

The comparison between Tables III and IV underscores the significant improvements achieved by using FinGPT-generated datasets for training machine learning models in credit risk assessment. The Random Forest model, which consistently outperforms the Decision Tree and K-Nearest Neighbors models, serves as the benchmark for this analysis.

In Table III, where models were trained on the original dataset, the Random Forest model demonstrates robust predictive capabilities, particularly as the data percentage increases. This indicates that the model effectively captures the inherent patterns within the original data, especially when more data is available.

However, the results in Table IV reveal that the Random Forest model's performance is further enhanced when trained on the synthetic dataset generated by FinGPT. This improvement suggests that the generated data not only complements the original dataset but may also introduce diverse patterns and variations that improve the model's generalization abilities. The FinGPT-generated data seems to provide a richer training set, which enables the Random Forest model to make more accurate predictions, even when compared to its performance on the original data.

The superior outcomes observed with the generated dataset indicate that FinGPT is highly effective in creating synthetic data that enhances the predictive accuracy of credit risk models. This is particularly beneficial for financial institutions with limited access to large, high-quality datasets. The ability of FinGPT to produce synthetic data that boosts model performance highlights its potential as a valuable tool in overcoming the limitations associated with small or imbalanced datasets, ultimately leading to more reliable and accurate credit risk assessments.

However, despite FinGPT's significant advantages in data generation, there are still some limitations. For example, synthetic data may not fully capture the complex nonlinear relationships present in real financial scenarios, or the quality of the generated data may be constrained by the quality of the original data. Additionally, relying on synthetic data could lead to model overfitting, which may impact the model's generalization ability in practical applications. Moreover, exploring the integration of FinGPT with other advanced data augmentation techniques such as Generative Adversarial Networks [19-21], to address a wider range of application scenarios may be a direction worth considering in the future.

IV. CONCLUSION

This study addressed the challenge of limited and imbalanced datasets in credit risk assessment by leveraging FinGPT, an open-source large language model, to generate synthetic datasets. This study aimed to enhance the predictive accuracy of machine learning models by training them on both original and FinGPT-generated datasets.

The results demonstrated that models trained on synthetic data outperformed those trained on traditional datasets, particularly in cases where data was scarce. For instance, the Random Forest model trained on synthetic data achieved higher accuracy (0.8903) and F1 score (0.8713) compared to the original dataset (accuracy 0.8654, F1 score 0.8504). This indicates that FinGPT-generated data enriched model performance by introducing diverse patterns do not present in the original data.

However, some limitations still remain. The synthetic data may not fully capture the complexity of real-world financial environments, potentially affecting generalizability in more dynamic settings. Additionally, only three machine learning models were tested, limiting the scope of the findings.

Future work will focus on improving the quality of FinGPT-generated data and exploring its application to more advanced machine learning models and broader financial tasks, such as fraud detection. Expanding FinGPT's role in financial modeling could further address data limitations and enhance decision-making accuracy in financial institutions.

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